

Tables 1–13 and Appendix Tables

Unifying Fairness Privacy Robustness and Explainability in Federated
Graph Neural Networks for High Stakes Risk Detection

Table 1: Where methods sit across the trust axes. $\checkmark$: supported; $\sim$: partial/adaptable; blank: not addressed. “Graph” = graph-native; “Group fair” = demographic group fairness (not merely client fairness); “Crypto” = validated on a large-scale cryptocurrency graph. [‡]Our DP entry is *not* equivalent to the record-level guarantees of DP-FedAvg, PUFFLE, FDP-Fair or FaVGNN: TrustFedGNN privatises the *released fairness statistic* rather than the whole model update (Section ??), so this column marks the presence of a formal (ϵ, δ) guarantee, not guarantees of equal scope.

Method	Fed.	Graph	Group fair	DP	Byz.	Explain	Crypto
FairGNN [?]		$\checkmark$	$\checkmark$			$\sim$	
NIFTY [?]		$\checkmark$	$\checkmark$				
FairSIN [?]		$\checkmark$	$\checkmark$				
DAB-GNN [?]		$\checkmark$	$\checkmark$				
FairFed [?]	$\checkmark$		$\checkmark$				
q-FedAvg [?]	$\checkmark$		$\sim$				
PUFFLE [?]	$\checkmark$		$\checkmark$	$\checkmark$			
FedFACT [?]	$\checkmark$		$\checkmark$				
DP-FedAvg [?]	$\checkmark$			$\checkmark$			
Krum/Median [? ?]	$\checkmark$				$\checkmark$		
F ² GNN [?]	$\checkmark$	$\checkmark$	$\checkmark$				
FairGFL [?]	$\checkmark$	$\checkmark$	$\sim$	$\checkmark$			
FaVGNN [?]	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$			
FDP-Fair [?]	$\checkmark$		$\checkmark$	$\checkmark$			
FedGraph-Fair [?]	$\checkmark$	$\checkmark$	$\checkmark$				
PoPETs’25 [?]	$\checkmark$		$\checkmark$	$\checkmark$			
TrustFedGNN (ours)	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark^{\ddagger}$	$\checkmark$	$\checkmark$	$\checkmark$

Table 2: Datasets. [†]Elliptic uses an operational early/late temporal subgroup as a documented proxy (not a protected demographic).

Dataset	Nodes	Edges	Feat.	Sensitive attr.	Pos. rate	Task
German	1,000	43,484	26	Gender	0.70	credit risk
Bail	18,876	623,740	17	Race (WHITE)	0.38	recidivism
Credit	30,000	2,843,716	12	Age	0.78	default
Pokec-z	67,796	1,235,916	276	Region	0.08	working field
Elliptic	203,769	234,355	165	Time period [†]	0.02	illicit (crypto)

Table 3: Paired Wilcoxon signed-rank tests (TrustFedGNN vs. each baseline over shared seeds): p -values *rounded to two decimals*, * marks $p < 0.05$; where the running text quotes a p -value it quotes this rounded value. “–” where fewer than five shared seeds are available (Credit/Elliptic remain at 2–3 seeds by cost, and are run against a restricted baseline set—see Section ??). Note the Wilcoxon floor: at $n=5$ (Bail, Pokec) the smallest attainable p is 0.0625, so a uniform 0.06 indicates our method wins on *all five* paired seeds, not a weak effect. On German (10 seeds) TrustFedGNN’s DPD is *not* significantly different from the non-fair backbone or most fair baselines; its AUC is significantly higher than the other (collapsing) privacy methods and lower than the non-private ones. On Bail its AUC beats every baseline on all seeds while its DPD ties the fairest non-collapsed methods; on Pokec it beats the backbone on *both* axes on all seeds.

Metric	Baseline	German	Bail	Credit	Pokec-z	Elliptic
AUC	FedAvg-GAT	0.02*	0.06	–	0.31	–
AUC	FairGNN	0.16	0.06	–	0.06	–
AUC	FairSIN	0.02*	0.06	–	0.19	–
AUC	FairFed	0.01*	0.06	–	0.06	–
AUC	F ² GNN	0.02*	0.06	–	0.06	–
AUC	FDP-Fair (2026)	0.01*	0.06	–	0.06	–
AUC	PUFFLE	0.00*	0.06	–	0.06	–
AUC	DP-FedAvg	0.01*	0.06	–	0.06	–
DPD	FedAvg-GAT	0.92	0.62	–	0.06	–
DPD	FairGNN	0.49	0.31	–	0.06	–
DPD	FairSIN	0.49	0.81	–	0.06	–
DPD	FairFed	0.01*	0.12	–	0.31	–
DPD	F ² GNN	0.11	0.81	–	0.44	–
DPD	FDP-Fair (2026)	0.00*	0.06	–	0.06	–
DPD	PUFFLE	0.01*	0.06	–	0.06	–
DPD	DP-FedAvg	0.01*	0.06	–	0.12	–

Table 4: Utility (AUC-ROC, higher is better). Mean $\pm$ std over seeds: **10** on the small/high-variance German, **5** on Bail and Pokec, and 2–3 on the larger Credit/Elliptic (see Table 3 for paired tests). Best per column in bold. TrustFedGNN is the only method that is fair *and* private (methods above the rule are non-private). *FaVGNN is a horizontal adaptation of a vertical-FL method. FairGFL/FedGraph-Fair/PUFFLE/FedFACT/PoPETs’25 are 2025–2026 reimplementations with approximated fidelity (Appendix ??).

Method	German	Bail	Credit	Pokec-z	Elliptic
FedAvg-GCN	0.715 $\pm$ 0.040	0.963 $\pm$ 0.006	0.762 $\pm$ 0.005	0.761 $\pm$ 0.004	–
FedAvg-GAT	0.708 $\pm$ 0.052	0.972 $\pm$ 0.001	0.765 $\pm$ 0.000	0.774 $\pm$ 0.010	0.970 $\pm$ 0.001
FairGNN	0.668 $\pm$ 0.084	0.906 $\pm$ 0.060	0.727 $\pm$ 0.001	0.703 $\pm$ 0.022	–
FairSIN	0.689 $\pm$ 0.043	0.956 $\pm$ 0.003	0.761 $\pm$ 0.002	0.762 $\pm$ 0.016	0.969 $\pm$ 0.001
FairFed	0.715 $\pm$ 0.045	0.961 $\pm$ 0.004	0.755 $\pm$ 0.000	0.756 $\pm$ 0.013	–
q-FedAvg	0.712 $\pm$ 0.051	0.961 $\pm$ 0.007	0.766 $\pm$ 0.002	0.762 $\pm$ 0.016	–
F ² GNN	0.697 $\pm$ 0.043	0.971 $\pm$ 0.004	0.753 $\pm$ 0.000	0.765 $\pm$ 0.010	–
FaVGNN* (2026)	0.626 $\pm$ 0.071	0.870 $\pm$ 0.101	–	0.640 $\pm$ 0.118	–
FDP-Fair (2026)	0.534 $\pm$ 0.062	0.552 $\pm$ 0.114	–	0.613 $\pm$ 0.042	–
FairGFL (2025)	0.708 $\pm$ 0.042	0.963 $\pm$ 0.006	–	0.764 $\pm$ 0.005	–
FedGraph-Fair (2026)	0.689 $\pm$ 0.049	0.964 $\pm$ 0.007	–	0.759 $\pm$ 0.006	–
PUFFLE	0.513 $\pm$ 0.076	0.606 $\pm$ 0.097	–	0.561 $\pm$ 0.055	–
FedFACT (2025)	0.706 $\pm$ 0.051	0.963 $\pm$ 0.006	–	0.758 $\pm$ 0.005	–
PoPETs’25 FairFed	0.707 $\pm$ 0.041	0.965 $\pm$ 0.005	–	0.761 $\pm$ 0.009	–
DP-FedAvg	0.523 $\pm$ 0.070	0.587 $\pm$ 0.119	0.481 $\pm$ 0.040	0.615 $\pm$ 0.043	–
TrustFedGNN (no DP)	0.634 $\pm$ 0.028	0.990 $\pm$ 0.004	0.752 $\pm$ 0.008	0.791 $\pm$ 0.011	0.943 $\pm$ 0.003
TrustFedGNN	0.628 $\pm$ 0.039	0.988 $\pm$ 0.006	0.747 $\pm$ 0.013	0.789 $\pm$ 0.016	0.944 $\pm$ 0.004
TrustFedGNN-Robust	0.667 $\pm$ 0.018	0.985 $\pm$ 0.005	0.760 $\pm$ 0.003	0.791 $\pm$ 0.005	0.946 $\pm$ 0.003

Table 5: Demographic Parity Difference (DPD, lower is fairer).

Method	German	Bail	Credit	Pokec-z	Elliptic
FedAvg-GCN	0.075 ± 0.056	0.065 ± 0.006	0.064 ± 0.003	0.033 ± 0.007	–
FedAvg-GAT	0.080 ± 0.060	0.070 ± 0.005	0.065 ± 0.002	0.034 ± 0.007	0.061 ± 0.006
FairGNN	0.088 ± 0.064	0.047 ± 0.028	0.104 ± 0.008	0.039 ± 0.014	–
FairSIN	0.067 ± 0.044	0.065 ± 0.007	0.062 ± 0.007	0.032 ± 0.008	0.038 ± 0.011
FairFed	0.027 ± 0.021	0.054 ± 0.009	0.003 ± 0.003	0.008 ± 0.006	–
q-FedAvg	0.085 ± 0.047	0.068 ± 0.005	0.064 ± 0.005	0.032 ± 0.006	–
F ² GNN	0.038 ± 0.028	0.064 ± 0.005	0.007 ± 0.001	0.010 ± 0.009	–
FaVGNN* (2026)	0.069 ± 0.054	0.049 ± 0.016	–	0.027 ± 0.024	–
FDP-Fair (2026)	0.006 ± 0.005	0.001 ± 0.000	–	0.001 ± 0.001	–
FairGFL (2025)	0.080 ± 0.052	0.068 ± 0.005	–	0.033 ± 0.006	–
FedGraph-Fair (2026)	0.111 ± 0.043	0.067 ± 0.004	–	0.032 ± 0.005	–
PUFFLE	0.011 ± 0.007	0.001 ± 0.001	–	0.003 ± 0.002	–
FedFACT (2025)	0.077 ± 0.062	0.021 ± 0.014	–	0.024 ± 0.010	–
PoPETs’25 FairFed	0.073 ± 0.052	0.066 ± 0.005	–	0.033 ± 0.005	–
DP-FedAvg	0.010 ± 0.011	0.001 ± 0.001	0.009 ± 0.005	0.004 ± 0.002	–
TrustFedGNN (no DP)	0.040 ± 0.035	0.070 ± 0.013	0.015 ± 0.009	0.012 ± 0.009	0.014 ± 0.014
TrustFedGNN	0.079 ± 0.059	0.066 ± 0.013	0.037 ± 0.014	0.013 ± 0.005	0.039 ± 0.008
TrustFedGNN-Robust	0.081 ± 0.086	0.058 ± 0.008	0.026 ± 0.002	0.014 ± 0.007	0.003 ± 0.002

Table 6: Equal Opportunity Difference (EOD, lower is fairer).

Method	German	Bail	Credit	Pokec-z	Elliptic
FedAvg-GCN	0.087 ± 0.073	0.021 ± 0.007	0.097 ± 0.028	0.033 ± 0.023	–
FedAvg-GAT	0.081 ± 0.067	0.024 ± 0.008	0.096 ± 0.025	0.023 ± 0.008	0.020 ± 0.010
FairGNN	0.058 ± 0.063	0.017 ± 0.012	0.117 ± 0.011	0.013 ± 0.009	–
FairSIN	0.074 ± 0.063	0.025 ± 0.013	0.103 ± 0.028	0.018 ± 0.010	0.019 ± 0.012
FairFed	0.046 ± 0.024	0.018 ± 0.010	0.072 ± 0.019	0.032 ± 0.020	–
q-FedAvg	0.095 ± 0.055	0.024 ± 0.011	0.105 ± 0.033	0.038 ± 0.026	–
F ² GNN	0.047 ± 0.038	0.021 ± 0.009	0.073 ± 0.010	0.037 ± 0.022	–
FaVGNN* (2026)	0.084 ± 0.088	0.020 ± 0.008	–	0.016 ± 0.008	–
FDP-Fair (2026)	0.120 ± 0.107	0.006 ± 0.004	–	0.014 ± 0.009	–
FairGFL (2025)	0.071 ± 0.049	0.028 ± 0.008	–	0.030 ± 0.013	–
FedGraph-Fair (2026)	0.090 ± 0.041	0.022 ± 0.009	–	0.036 ± 0.024	–
PUFFLE	0.102 ± 0.088	0.006 ± 0.013	–	0.023 ± 0.013	–
FedFACT (2025)	0.097 ± 0.077	0.012 ± 0.007	–	0.088 ± 0.041	–
PoPETs'25 FairFed	0.072 ± 0.046	0.023 ± 0.008	–	0.022 ± 0.020	–
DP-FedAvg	0.139 ± 0.090	0.012 ± 0.008	0.108 ± 0.083	0.011 ± 0.007	–
TrustFedGNN (no DP)	0.061 ± 0.062	0.015 ± 0.015	0.065 ± 0.006	0.024 ± 0.011	0.058 ± 0.022
TrustFedGNN	0.079 ± 0.078	0.009 ± 0.007	0.068 ± 0.012	0.019 ± 0.013	0.075 ± 0.021
TrustFedGNN-Robust	0.087 ± 0.096	0.005 ± 0.004	0.083 ± 0.022	0.021 ± 0.017	0.079 ± 0.013

Table 7: Ablation of TrustFedGNN components (no DP). AUC / DPD per dataset. The ablation is a *separate* set of runs from the main comparison (its own matched seed set, so that all four configurations are compared on identical seeds); its numbers are therefore close to, but not identical with, the corresponding rows of Tables 4 and 5, which aggregate the full seed sets (10 on German, 5 on Bail). All four differences lie within one standard deviation of the multi-seed mean—e.g. German DPD 0.045 here versus 0.040 ± 0.035 there, and Bail AUC 0.988 versus 0.990 ± 0.004 —so the comparison within this table is internally consistent even where a cell does not reproduce the main table exactly.

Config	Bail		German	
	AUC	DPD	AUC	DPD
Base GAT	0.972	0.072	0.709	0.050
+ BFWA (no FSER)	0.972	0.073	0.680	0.071
+ FSER (no BFWA)	0.987	0.059	0.668	0.009
+ FSER + BFWA	0.988	0.075	0.638	0.045

Table 8: Attribute-inference attack on the released fairness statistic (Bail, real predictions). A worst-case differencing adversary infers a target node’s sensitive attribute. The exact release is a perfect oracle; FTGD’s noise reduces inference to balanced-random (0.500) at every deployed ϵ . Targets are sampled in equal numbers from each true group (2,000 each), so 0.500—not the dataset’s 0.53 majority-group prevalence—is the correct chance reference.

Release	ϵ	Noise mult. z	Inference acc.
Exact (s -statistic, no DP)	∞	0	1.000
FTGD	16	3.6	0.502
FTGD	8	6.4	0.510
FTGD	4	11.6	0.506
FTGD	2	21.5	0.513
FTGD	1	40.5	0.507
FTGD	0.5	76.7	0.503
<i>Random guess (balanced targets)</i>	—	—	0.500

Table 9: Utility *and* fairness retained under attack (2/10 Byzantine clients, Bail, single seed). AUC (higher = more robust), DPD and EOD (lower = fairer). Best per attack-metric in bold among non-collapsed models; [†] marks near-chance models whose low disparity is an artefact of constant predictions.

Aggregator	Gaussian			ALIE			Fair-poison		
	AUC $\uparrow$	DPD $\downarrow$	EOD $\downarrow$	AUC $\uparrow$	DPD $\downarrow$	EOD $\downarrow$	AUC $\uparrow$	DPD $\downarrow$	EOD $\downarrow$
fedavg	0.511	0.031 [†]	0.048 [†]	0.935	0.062	0.039	0.945	0.015	0.037
bfwa	0.483	0.014 [†]	0.006 [†]	0.991	0.086	0.030	0.689	0.086	0.126
krum	0.986	0.053	0.022	0.710	0.032	0.003	0.991	0.049	0.007
multikrum	0.985	0.060	0.017	0.916	0.063	0.030	0.989	0.068	0.024
median	0.962	0.066	0.033	0.790	0.051	0.020	0.980	0.038	0.010
trimmed_mean	0.984	0.055	0.018	0.956	0.060	0.035	0.983	0.049	0.005
robust_bfwa (ours)	0.992	0.075	0.018	0.953	0.045	0.034	0.990	0.068	0.024

Table 10: Million-node scaling study (ogbn-products, 2.4M nodes, 123M edges, $K=3$ clients, 15 rounds, CPU). *2026 baseline. [†]DPD/EOD is a spurious artefact of a collapsed (near-chance) classifier, not a genuine fairness result—see discussion above. The energy column is a transparent *proxy*, not a metered measurement: logged wall-clock $\times$ an assumed 65 W CPU draw (`src/trust/sustainability.py`), valid for relative comparison between methods on this one machine and not comparable across hardware.

Method	AUC $\uparrow$	DPD $\downarrow$	EOD $\downarrow$	Time/run (s)	Energy (Wh)
FedAvg-GAT	0.877	0.101	0.039	1449	26.2
FairSIN	0.894	0.095	0.011	1345	24.3
FaVGNN* (2026)	0.495	0.000 [†]	0.000 [†]	2791	50.4
DP-FedAvg	0.556	0.009 [†]	0.021 [†]	1251	22.6
TrustFedGNN (no DP)	0.861	0.114	0.071	3550	64.1
TrustFedGNN	0.862	0.103	0.053	3665	66.2

Table 11: Calibration, uncertainty, and efficiency (Bail). ECE and Brier (lower better), group uncertainty gap, and per-round communication. Energy is *not* reported here: wall-clock was not logged for these canonical small-graph runs, so the proxy of Section ?? has no input for them. We report it instead where the measurement exists, on the large-scale study (Table 10), rather than print an uninformative zero.

Method	ECE	Brier	U-gap	Comm/rd (MB)	Trust
FedAvg-GAT	0.031	0.054	0.002	0.225	0.650
DP-FedAvg	0.322	0.274	0.002	0.215	0.480
TrustFedGNN (no DP)	0.043	0.061	0.002	0.896	0.632
TrustFedGNN	0.062	0.073	0.006	0.896	0.718

Table 12: Composite trust score on Bail: utility, fairness, and privacy combined via the arithmetic-mean ($p=1$) power mean, from the same canonical runs as Tables 4–6. Non-private methods score 0 on privacy by construction. The calibration-augmented composite is in Table 11. Ranking is sensitive to the reference constants (stated in text); best in bold.

Method	AUC	DPD	EOD	ϵ	Trust
FedAvg-GAT	0.972	0.070	0.024	–	0.579
FairGNN	0.906	0.047	0.017	–	0.582
FairSIN	0.956	0.065	0.025	–	0.577
FairFed	0.961	0.054	0.018	–	0.593
F ² GNN	0.971	0.064	0.021	–	0.586
DP-FedAvg	0.587	0.001	0.012	8	0.685
TrustFedGNN (no DP)	0.990	0.070	0.015	–	0.593
TrustFedGNN	0.988	0.066	0.009	8	0.767

Table 13: Mapping of TrustFedGNN components to EU AI Act articles and NIST AI RMF functions.

Requirement	Addressed by
<i>EU AI Act</i>	
Art. 10 - Data & data governance	Non-IID federated silos keep raw data local; sensitive attribute excluded from features; documented dataset provenance and base-rate gaps.
Art. 13 - Transparency & information	Per-edge FSER attention and feature-level disparity attribution expose why a decision is made; model card documents intended use and metrics.
Art. 14 - Human oversight	Explicit, human-set fairness budget τ and privacy budget ϵ ; Pareto trade-off curves let an operator choose the operating point.
Art. 15 - Accuracy, robustness & cybersecurity	Byzantine-robust aggregation (robust_bfwa) with reported accuracy retention under poisoning/fairness attacks; RDP privacy guarantee.
<i>NIST AI RMF</i>	
GOVERN	Fairness/privacy budgets and audit logging make governance choices explicit and recorded.
MAP	Threat model (Byzantine + fairness poisoning) and subgroup-disparity context are documented.
MEASURE	DPD/EOD/EO, RDP ϵ , attack-retention, ECE and a composite trust score are all measured.
MANAGE	Robust aggregation + fairness constraint actively mitigate measured risks each round.

Table 14: Hyperparameters. “Explored” gives the sweep range where applicable.

Hyperparameter	Default	Explored
GNN layers L	2	$\{2, 3\}$
Hidden dimension d_h	64	$\{32, 64, 128\}$
Attention heads (FSER/GAT)	4	$\{1, 4\}$
Dropout	0.3	$\{0.2, 0.3, 0.5\}$
Local optimiser	AdamW	—
Local learning rate η	0.01	$\{10^{-3}, 10^{-2}\}$
Weight decay	10^{-5}	—
Local epochs E	2	$\{1, 2, 3\}$
Communication rounds T	40–60	per dataset
Clients K	3–10	$\{3, 5, 10, 20\}$
Partition	Dirichlet($\alpha=0.5$)	$\alpha \in \{0.1, 0.5, 1.0\}$
Fairness weight λ	1.0	$\{0, 0.25, 0.5, 1, 2, 4\}$
Fairness budget τ	0.05	$\{0.01, 0.05, 0.1\}$
DP ϵ	8	$\{0.5, 1, 2, 4, 8, 16\}$
DP δ	10^{-5}	—
DP clip C	1.0	—
Frank–Wolfe iters T_{FW}	20	—
Dual step η_μ	0.1	—
MC-dropout samples	20	—

Table 15: What each 2025–2026 baseline’s reimplementation keeps vs. drops.

Baseline	Kept	Dropped / approximated
FairGFL	weight $\propto 1/(1+O_i)$ aggregation rule	O_i (multi-graph node/edge overlap) proxied by sample-count deviation from the mean, since our setting has one Dirichlet-partitioned graph, not separate per-client graphs
FedGraph-Fair	minimax/DRO dual λ reweighting high-loss clients	personalised per-client models mixed via a learned peer-similarity graph (orthogonal decentralised-communication contribution)
PUFFLE	momentum controller auto-tuning the fairness weight toward a target disparity, + DP-SGD	the third privacy channel (noised group-count sharing for clients missing a demographic group) – unneeded here since every client observes both groups
FedFACT	closed-form global+local offset (exact reduction of the paper’s optimum for binary demographic parity)	general multiclass cost-matrix calibration and the iterative dual-ascent solver
PoPETs’25	degree-2-polynomial FairFed weighting (the FHE-friendly surrogate itself)	threshold-CKKS secure aggregation and its noise-as-DP analysis (cryptographic infrastructure, no numeric effect once computed in the clear)